



Metabolic Changes of Perimenopause and Nutritional Advice

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Received: 13 May 2026 / Accepted: 3 August 2026
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Abstract

Purpose of Review Perimenopause is a period of hormonal volatility associated with clinically relevant changes in metabolism and cardiometabolic risk. In addition to vasomotor symptoms, sleep disruption, and changes in body composition, the transition from the late reproductive years to early postmenopause is characterized by physiologic alterations that may precede formal diagnostic thresholds for metabolic syndrome. However, screening and risk stratification are often anchored to postmenopausal benchmarks, potentially delaying identification of individuals at elevated risk.

Recent Findings Accumulating evidence supports earlier attention to dynamic indices of cardiometabolic deterioration during perimenopause. Lifestyle interventions and, for selected individuals, menopausal hormone therapy may mitigate progression, although important uncertainties remain, including the long-term effectiveness of precision nutrition approaches and genotype-informed strategies.

Summary This narrative review summarizes physiologic changes across the menopausal transition and evaluates nutrition and lifestyle interventions that may attenuate adverse metabolic trajectories.

Keywords Perimenopause · Cardiometabolic health

Introduction

For nearly half a century, menopause has often been conceptualized in clinical and research settings as a discrete end point marked by the final menstrual period, after which cardiovascular and metabolic risk increases. While this framework is useful for epidemiologic categorization, it can obscure the earlier and more dynamic phase of perimenopause, during which symptoms emerge and cardiometabolic trajectories may begin to shift.

Clinical Definition of Menopause

The Stages of Reproductive Aging Workshop (STRAW + 10) describes the menstrual, reproductive, and hormonal changes that occur from the late reproductive stage through early postmenopause [1]. Collectively, these stages comprise the menopausal transition, commonly referred to as perimenopause [1]. Perimenopause typically spans four to eight years and is defined not by stable estrogen deficiency but by hormonal variability, including intermittent estradiol surges and progressive progesterone withdrawal. Emerging evidence suggests that this phase represents a distinct metabolic transition during which risk for subsequent dysglycemia, steatotic liver disease, and cardiovascular disease may increase, including in the absence of overt clinical symptoms [2].

The late reproductive stage (LRS) is associated with endocrinologic changes, including declining inhibin B and antimüllerian hormone, which contribute to reduced fecundability. In the years preceding the final menstrual period (FMP), perimenopause is characterized by fluctuations in estradiol, progesterone, and follicle stimulation hormone (FSH) that contribute to progressive changes in menstrual cycle pattern [3–5]. Despite these general patterns,

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substantial interindividual variability is observed [3]. Disruption of hypothalamic–ovarian axis feedback mechanisms can lead to cycle-to-cycle variability in luteinizing hormone (LH), FSH, estradiol, and progesterone concentrations [3, 6]. Menopause is defined retrospectively, after 12 consecutive months without menses, reflecting the end of follicular ovarian function. The mean age at menopause is 51.4 years, with approximately 90% of individuals reaching this milestone between ages 45 and 56 years [7, 8]. Symptoms associated with the menopausal transition often begin before the FMP and persist for a median duration of 7.4 years [9].

Physical Symptoms of the Perimenopause

The menopausal transition is associated with symptoms across multiple organ systems [1, 9]. Central nervous system manifestations include vasomotor symptoms (hot flashes and night sweats), which are thought to arise in part from altered hypothalamic thermoregulatory control. Fluctuating estrogen levels may modify signaling in thermoregulatory centers, narrowing the thermoneutral zone [9]. Estrogen also modulates neurotransmitters such as serotonin and gamma-aminobutyric acid (GABA) and is implicated in cerebral energy metabolism; variability in sex steroid concentrations has been associated with mood symptoms, sleep disruption, cognitive complaints, and changes in sexual health [9]. Declining estrogen contributes to changes in body composition associated with bothersome weight gain [4, 9]. Estrogen withdrawal is associated with accelerated bone resorption and reduced bone density, loss of lean muscle mass, and progression of osteoarthritic symptoms [9, 10]. Fluctuating estradiol levels may contribute to hypothalamic thermoregulatory instability, narrowing the thermoneutral zone and promoting vasomotor symptoms that fragment sleep [11, 12]. Declining estrogen and rising FSH levels have been associated with impaired sleep quality and reduced sleep efficiency [13, 14].

Decreased collagen production and alterations in estrogen/androgen ratios may contribute to hair and skin changes. Finally, urogenital tissues are highly estrogen sensitive, and reduced estrogen is associated with progressive vaginal dryness, dyspareunia, urinary symptoms, and increased susceptibility to urinary tract infections [9]. Sleep disorders such as insomnia are common during perimenopause and are associated with reduced quality of life, fatigue and impaired work functioning [11, 13]. The longitudinal Study of Women's Health Across the Nation (SWAN) estimated the risk of sleep problems in perimenopausal women to be higher by a factor of 1.5 compared with premenopausal women [15]. Sleep disruption continues to persist into menopause reported by 16% to 47% of perimenopausal women, with

prevalence increasing to 35% to 60% after menopause [11, 16]. In observational studies, sleep disruption is also associated with worsening mood symptoms, impaired attention, and subjective memory concerns [13]. Chronic sleep disruption has been linked to weight gain, obesity, metabolic dysfunction, type 2 diabetes, and cardiovascular disease in broader populations [17, 18].

Physiologic Basis of Perimenopausal Symptoms and Metabolic Risk

Hormonal Nexus: Estrogen, Inflammation, and Adipose Dysfunction

A central feature of perimenopause is hormonal volatility, which has been associated with metabolic changes that may not be fully captured by experimental models of fixed estrogen deprivation. During this period, estrogen receptor- α (ER α) signaling in skeletal muscle, liver, and adipose tissue may become less responsive, contributing to a state of relative hormone resistance that can precede sustained estrogen deficiency. In parallel, declining progesterone may reduce anti-inflammatory signaling, potentially favoring expansion of visceral adipose tissue even in the absence of large changes in total body weight. Hypertrophied visceral adipocytes increase delivery of free fatty acids to the portal circulation and may promote macrophage polarization toward a more proinflammatory phenotype, contributing to elevations in interleukin-6 and tumor necrosis factor- α [19]. This low-grade inflammatory milieu may, in turn, reinforce visceral adiposity and impair insulin signaling through multiple pathways.

Deterioration of Systemic Metabolism: Energy Expenditure and Metabolic Inflexibility

Longitudinal data from several prospective cohorts have consistently demonstrated that the perimenopausal transition is accompanied by a steepening decline in insulin-stimulated glucose disposal, an effect that remains significant after adjustment for chronological age and body mass index [20, 21]. Proposed mechanisms include reduced glycogen synthesis in skeletal muscle, impaired suppression of hepatic glucose production, and progressive deterioration of pancreatic beta-cell function [22]. Fasting glucose often remains within the normal reference range until late perimenopause, whereas postprandial glucose excursions and fasting insulin levels may increase earlier. Consequently, reliance on isolated fasting glucose measurements may underestimate the burden of dysglycemia during this phase. Dynamic measures such as the homeostatic model assessment of insulin

resistance (HOMA-IR) have been proposed as more sensitive markers in at-risk perimenopausal women, even when fasting glucose is unremarkable [23, 24]. Collectively, these observations suggest that earlier use of dynamic glycemic measures could improve risk stratification and identify dysglycemia before diagnostic thresholds for prediabetes or diabetes are met. There are not yet consensus or best practices for use of these tools, or how they should be integrated into clinical practice.

Beyond glucose dysregulation, the perimenopausal transition is associated with a shift toward a more atherogenic lipid profile. Total cholesterol, low-density lipoprotein cholesterol, and triglycerides tend to increase, while high-density lipoprotein cholesterol declines, patterns that appear to persist after adjustment for age and adiposity [4, 25]. Proposed mechanisms include estrogen-dependent changes in hepatic lipase and lipoprotein lipase activity, as well as increased flux of free fatty acids from visceral adipose tissue. Untargeted metabolomic studies have also reported a signature of elevated branched-chain amino acids (leucine, isoleucine, valine) and aromatic amino acids (phenylalanine, tyrosine) emerging during perimenopause and persisting into postmenopause [26, 27]. Elevated branched-chain amino acids have been associated with future insulin resistance and type 2 diabetes independent of traditional risk factors [28]. This amino acid profile correlates with *de novo* lipogenesis in the liver and may represent an early biomarker of metabolic deterioration during the transition.

Resting metabolic rate declines with age. Whether the hormonal fluctuations of perimenopause independently affect energy metabolism—beyond the effects of chronological aging, changes in body composition, and lifestyle factors—remains debated. Several longitudinal studies suggest these changes are not fully explained by chronological aging or body composition alone; rather they reflect unique endocrine transitions with direct metabolic consequences. Mechanistic research points to disrupted estrogen receptor signaling, mitochondrial dysfunction, and neuroendocrine uncoupling as contributors to hypometabolic states during perimenopause [29, 30]. However, other studies suggest that menopausal hormonal changes may not exert a significant independent effect on resting energy expenditure after adjustment for age and body composition. For example, Karppinen et al. reported that adjusted resting energy expenditure declined by approximately 3.9 kcal/day per year of age, without clear evidence of an additional independent effect attributable to menopausal status [31].

In addition to the hormonal and metabolic changes, body composition also shifts. Redistribution of adiposity toward the visceral compartment has been reported, with visceral adipose tissue increasing by approximately 8.2% per year in the two years preceding the final menstrual period and

by 5.8% per year thereafter [32]. Cross-sectional pooled analyses similarly suggest that postmenopausal women have approximately 26.9 cm² greater visceral fat area and 4.63 cm greater waist circumference than premenopausal women [33].

In SWAN, lean mass changed from + 0.2% per year before to − 0.2% per year during the transition, representing an average loss of approximately 0.2 kg [34]. Bone mineral density also declines across the transition, with femoral neck bone loss accelerating from − 0.51% per year at ages 45–49 to − 1.39% per year at ages 50–54, and lumbar spine bone loss accelerating from − 0.56% to − 1.33% per year over the same interval [35]. Collectively, these data suggest that perimenopause is characterized less by weight gain than by central fat accumulation, modest loss of lean tissue, age-related metabolic slowing, and accelerated skeletal loss.

The cumulative effect of these systemic alterations is associated with higher prevalence of metabolic syndrome across the menopausal transition. A large systematic review and meta-analysis of observational studies reported a pooled prevalence of metabolic syndrome in postmenopausal women of 37.2% (95% confidence interval 35.0–39.3%), corresponding to a 1.5- to 4-fold higher risk compared with premenopausal women [36]. In observational datasets, changes occur across all five components of the syndrome (waist circumference, triglycerides, blood pressure, fasting glucose, and high-density lipoprotein cholesterol). Mechanistically, estrogen withdrawal has been linked to activation of the renin-angiotensin-aldosterone system and increased vascular smooth muscle tone, which may contribute to higher blood pressure independent of obesity [37]. Consistent with this physiology, markers of cardiovascular disease risk—including coronary artery calcification, carotid intima-media thickness, and incident heart failure with preserved ejection fraction—have been reported to increase during the perimenopausal years in some cohorts [4, 8, 25, 38–41].

Cerebral Metabolism and Neuroendocrine Integration

The metabolic consequences of perimenopause may extend to the central nervous system. As previously described, estrogen and progesterone impact multiple neurohormonal signaling pathways. Some frameworks distinguish chronological from endocrinological influences on brain aging, with perimenopause proposed as a period during which these processes intersect. Positron emission tomography studies have reported reductions in cerebral glucose utilization during perimenopause, particularly in the hypothalamus, hippocampus, and prefrontal cortex—regions that are rich in estrogen receptors and participate in metabolic and

cognitive integration [42]. Reduced brain glucose metabolism has been described alongside microglial activation and elevated central proinflammatory cytokines, a profile that could plausibly contribute to fatigue, sleep disruption, and cognitive symptoms reported during the transition [43–45]. The hypothalamus integrates peripheral metabolic signals (e.g., leptin, insulin, ghrelin) with neuroendocrine outputs, including gonadotropin-releasing hormone and corticotropin-releasing hormone [46, 47]. Disruption of hypothalamic signaling during perimenopause has been hypothesized to influence appetite regulation, visceral adiposity, and cortisol dynamics, which may contribute to impaired peripheral insulin sensitivity.

Nutritional Strategies to Address Perimenopausal Metabolic Changes

The hormonal and metabolic shifts of perimenopause are associated with changes relevant to skeletal health, muscle mass, gastrointestinal physiology, and cardiometabolic risk. Nutritional strategies have therefore been examined as potential adjuncts to other interventions across the menopausal transition. The following section summarizes evidence and guideline-based considerations for selected dietary components and supplements that have been studied in relation to perimenopausal and early postmenopausal outcomes.

Dietary Strategies

Recognition that perimenopause is an active, hormonally mediated metabolic state has implications for prevention and treatment strategies. Lifestyle interventions are commonly recommended as initial management; however, intervention components and expected response may differ across the menopausal transition [48]. The Mediterranean dietary pattern is among the most consistently studied nutritional approaches during menopause transition. An European Menopause and Andropause Society (EMAS) position statement synthesizing observational and interventional evidence concluded that adherence to a Mediterranean diet may improve vasomotor symptoms, mood, and depressive symptoms in midlife women, potentially through anti-inflammatory effects, improved insulin sensitivity, and favorable neurovascular modulation [49].

Some studies suggest that caloric restriction alone may be insufficient to improve metabolic flexibility in all individuals, whereas combined approaches incorporating modest energy deficit (approximately 300–500 kcal/day), dietary patterns higher in unsaturated fats and fiber, and adequate protein intake are associated with improvements in visceral adiposity and insulin sensitivity [50].

Intermittent fasting in menopause has been evaluated in a small number of clinical studies, mostly using time-restricted eating. A recent quasi-randomized controlled trial showed that combining time restricted eating (TRE) with exercise improved menopausal symptoms and quality of life compared to exercise alone, although vasomotor symptoms were unchanged [51]. Earlier controlled studies in menopausal women demonstrated that TRE plus exercise leads to greater reductions in body mass index (BMI) and improvements in body composition than exercise alone, while randomized and subgroup analyses indicate that weight loss and metabolic benefits of TRE are similar in pre- and postmenopausal women [52].

Protein Intake

Loss of lean mass during and after the menopausal transition has motivated interest in protein adequacy as a potentially modifiable factor. The European Society for Clinical and Economic Aspects of Osteoporosis and Osteoarthritis (ESCEO) and the European Society for Clinical Nutrition and Metabolism (ESPEN) recommend 1.0–1.2 g/kg body weight/day for healthy older adults, with intake distributed across meals (20–25 g of high-quality protein per meal) to support muscle protein synthesis [53, 54]. Leucine-rich protein sources (e.g., dairy, eggs, legumes, and fish) have been emphasized in expert guidance, and resistance exercise is consistently associated with augmentation of the anabolic response to dietary protein [54, 55]. While adequate protein is necessary, it is not sufficient to maintain or increase muscle mass and must be combined with adequate physical activity to increase muscle mass [56, 57]. Exercise interventions that combine resistance training (to preserve fat-free mass) with aerobic modalities, including higher-intensity interval training, have been associated with improvements in metabolic flexibility and postprandial lipid handling [58]. Overall, earlier initiation of lifestyle interventions during the perimenopausal years may be associated with larger effect sizes than interventions begun after metabolic syndrome is established, although direct comparative trials remain limited.

Dietary Fiber

Emerging gastroenterological evidence supports bidirectional relationships between gut physiology and perimenopausal metabolic dysregulation. A 2025 review described associations between menopause and changes in gut motility, microbiota composition, and intestinal barrier function, with potential downstream implications for systemic inflammation, gastrointestinal disease risk,

and cardiometabolic health [59, 60]. These gastrointestinal changes have been linked mechanistically to the estrobolome, the ensemble of gut bacterial genes encoding β -glucuronidase enzymes that deconjugate estrogens and facilitate enterohepatic reabsorption [25, 59]. Perturbations in this axis during perimenopause could plausibly influence circulating estrogen bioavailability and downstream metabolic risk, although causal pathways in humans remain an active area of investigation.

At a population level, higher dietary fiber intake has been associated with lower metabolic syndrome risk and mortality in postmenopausal cohorts. In National Health and Nutrition Examination Survey (NHANES) analyses, each 1 g/1,000 kcal increase in fiber density was associated with a 4% reduction in the odds of metabolic syndrome, with partial mediation through inflammatory markers [61]. In addition, fiber intake above 7.9 g/day was associated with lower all-cause and cardiovascular mortality [62]. Mechanistically, fiber fermentation supports microbial diversity relevant to estrobolome function, and short-chain fatty acids may reinforce intestinal barrier integrity, potentially countering menopause-associated gut barrier disruption [59]. Consistent with broader dietary guidance, recommended intakes are commonly cited in the range of 21–25 g/day from whole-food sources such as legumes, whole grains, vegetables, and fruit [63, 64].

Alcohol

Moderate alcohol consumption (10–14 g/day for women and 20–28 g/day for men or roughly 1 drink/day in women, 1–2 in men) has historically been considered a defining feature of the traditional Mediterranean diet—particularly when consumed with meals—but its inclusion is now under significant scientific scrutiny due to emerging evidence about associated health risks [65–67].

Alcohol consumption—especially at higher levels—is linked to poorer menopause-specific quality of life scores, particularly affecting vasomotor (hot flashes), psychosocial, and physical domains. Heavy drinking increases the risk of bothersome vasomotor symptoms, while light-to-moderate intake shows mixed or neutral associations [68, 69].

Across many large studies, even light to moderate alcohol intake is associated with a gradual increase in breast cancer risk in women [70]. The proposed mechanism by which alcohol heightened the risk for breast cancer in women is believed to be related to the alcohol increasing estradiol and testosterone and lowering sex hormone binding globulin, both in perimenopause and after menopause; these small hormonal shifts are thought to contribute to breast cancer risk [71].

Micronutrients

Calcium and Vitamin D

Accelerated bone loss during the menopausal transition underscores the importance of adequate calcium and vitamin D intake. Many guidelines recommend total calcium intake of approximately 1,000–1,200 mg/day (preferably from dietary sources) and vitamin D intake of 600–1,000 IU/day for postmenopausal women [72, 73]. Cholecalciferol (vitamin D3) is generally favored over ergocalciferol (D2) for raising and maintaining serum 25-hydroxyvitamin D concentrations [74]. Calcium plus vitamin D supplementation has been reported to confer modest benefits for bone mineral density in some populations, particularly among older women and those with high adherence; however, routine supplementation for all postmenopausal women is not uniformly recommended [72, 75]. When dietary calcium intake is insufficient, supplemental calcium may be used to meet total intake targets, with attention to limiting higher supplemental doses given associations with nephrolithiasis risk in some studies [75].

Vitamin K

Vitamin K2 (menaquinones) is involved in γ -carboxylation of osteocalcin and matrix Gla protein, processes that influence bone mineralization and may modulate vascular calcification pathways [76, 77]. A 2022 meta-analysis of 16 randomized controlled trials reported small improvements in lumbar spine bone mineral density with vitamin K2 supplementation (mean difference 1.02%, $p = 0.0006$), whereas evidence for fracture reduction was inconsistent across studies [76, 78]. In contrast, a 3-year randomized trial in osteopenic postmenopausal women found that menaquinone-7 (375 μ g/day) added to calcium and vitamin D did not prevent bone mineral density loss [79]. Notably, many of the trials reporting benefit were conducted in Japan, which may limit generalizability across populations and dietary contexts [78].

Magnesium

Magnesium has established roles in bone metabolism and vitamin D activation with 60% of it being stored in bone and notable impact on osteoclast and osteoblast activity, and notably many women have been found to consume less than the RDA [80–82]. While higher magnesium intake has been associated with higher bone mineral density (BMD), initial review of cohort studies did not find that higher magnesium intake was associated with reduced fracture risk [82]. More

recent studies have demonstrated that supplementation had demonstrated benefit in not only bone density but also reduced fracture risk across multiple populations, including women across through the menopausal transition, however the studies have been small in size [83]. Further larger clinical trials are needed to best determine optimal strategies to optimize skeletal needs for magnesium.

Taken together, available data support consideration of dietary patterns that address multiple physiologic domains relevant to the menopausal transition. While calcium and vitamin D targets are most directly linked to skeletal outcomes, additional dietary factors—including adequate high-quality protein intake and higher fiber consumption from diverse plant sources—may plausibly influence concurrent trajectories in muscle mass, gut-mediated estrogen metabolism, and cardiometabolic risk [25, 59]. Evidence for vitamin K2 and magnesium remains mixed and appears context dependent, underscoring the need for standardized trials in perimenopausal populations with clinically meaningful outcomes.

Additional Supplements

Interest in herbal and non-pharmacologic supplements for perimenopausal symptom management is common, but the evidence base remains heterogeneous and, for many agents, insufficient to support routine use. Creatine has been examined primarily in younger athletic populations and, in midlife women, more often in postmenopausal cohorts; long-term data specific to perimenopause remain limited [84].

Soy isoflavones act as selective estrogen receptor modulators and have been evaluated for menopausal symptoms, with effects that appear modest on average. These phytoestrogens are commonly used for vasomotor symptoms, however their benefit were not conclusively confirmed in a Cochrane analyses, although they found that genistein-rich soy extracts may reduce hot flashes in some women [85]. Across trials, heterogeneity is substantial, placebo responses are common, and benefits may attenuate in longer-duration studies [86]. Observational data from the InterLACE consortium suggest an association between frequent soy food consumption and lower likelihood of subsequent vasomotor symptoms [87]. Collectively, current evidence supports a potential symptomatic benefit in some individuals, although effect sizes are generally small and appear time dependent. Accordingly, for most supplements, conclusions are best framed as provisional pending adequately powered, standardized randomized trials with clinically meaningful end points.

Other Therapeutic Implications

Menopausal hormone therapy has a complex history in metabolic medicine, but evidence for initiation during the

perimenopausal window has become more favorable in some studies and may have metabolic implications. When initiated during this period, estrogen therapy (with or without a progestogen, as indicated) has been associated with reductions in visceral adipose accumulation, improvements in insulin sensitivity, lower fasting glucose, and attenuation of increases in blood pressure. Reported benefits appear larger among women who begin therapy before established insulin resistance or overt diabetes. However, hormone therapy is not a universal metabolic intervention: it does not reverse advanced hepatic fibrosis, and effects on body weight are generally modest, with some studies suggesting prevention of further gain rather than induction of loss [88, 89]. Clinical decision-making therefore typically weighs potential metabolic benefits against individual risks, including breast cancer, venous thromboembolism, and stroke. A post hoc analysis of the Women's Health Initiative suggested that baseline metabolic syndrome status may modify coronary outcomes, with coronary event risk significantly higher among women with metabolic syndrome who received hormone therapy, compared to women without metabolic syndrome [90]. These findings underscore the importance of risk stratification and individualized assessment.

Future Directions and Unresolved Questions

Understanding of perimenopause-associated metabolic change has advanced substantially; however, several questions remain unresolved. First, the optimal timing, duration, and formulation of hormone therapy for metabolic outcomes (as distinct from symptom relief) require dedicated randomized controlled trials with prespecified metabolic end points, rather than reliance on secondary analyses of cardiovascular outcome studies. Second, the potential role of genotype-informed precision nutrition during perimenopause remains largely unexplored. Polymorphisms in genes involved in estrogen metabolism (COMT, rs4680), one-carbon metabolism (MTHFR, rs1801133 and rs1801131), and fucosylation (FUT2, rs602662 and rs601338) have been associated with interindividual differences in weight gain, dyslipidemia, and insulin resistance, suggesting possible targets for personalized prevention strategies [91–94]. Third, the gut microbiome undergoes compositional shifts during perimenopause, including reduced diversity and an increase in proinflammatory taxa; whether microbiome-targeted interventions (e.g., prebiotics, probiotics, or fecal microbiota transplantation) can meaningfully alter metabolic trajectories remains uncertain. Fourth, associations between perimenopausal metabolic changes and later neurodegenerative outcomes, particularly in apolipoprotein E4 (APOE4) carriers, warrant prospective investigation. Finally, large-scale prospective studies that map tissue-specific estrogen signaling across

the perimenopausal years—integrating modern imaging, transcriptomic, and metabolomic approaches—are needed to define the earliest detectable alterations and identify optimal windows for intervention.

Conclusion

Perimenopause is a hormonally volatile state associated with a cascade of metabolic changes, including increasing visceral adiposity, declining insulin sensitivity, atherogenic dyslipidemia, hepatic steatosis, reductions in energy expenditure, and alterations in central neuroendocrine regulation. Many of these changes appear to precede clinical diagnosis by several years, highlighting perimenopause as both a window of vulnerability and a potential opportunity for earlier risk identification. Further work is needed to define optimal approaches to stage-specific screening and to determine which interventions most effectively attenuate adverse cardiometabolic trajectories when initiated during the perimenopausal years.

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 - Comprehensive review of cardiometabolic impacts during the menopausal transition. Describing not only physiologic change, but also discussion of treatment considerations to help mitigate cardiovascular risk.
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- Excellent review summarizing updated thinking on how hormone replacement therapy impacts the multiple hallmarks of aging pathways and should be reconsidered not only for symptom management but also as a geroprotector.

Author Contributions SL, HN, RS, DK, AM and EB all contributed to the conception and manuscript text. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Human and Animal Rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Consent for Publication This paper uses gendered terminology due to common terminology in most medical publications—however we recognize many patients with these concerns may identify differently.

Competing Interests The authors declare no competing interests.

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